

Лабораторна робота №5

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Група ІПЗ-21

Варіант 7

Код консольної програми

```
using System.Security.AccessControl;

class Program
{
    static void Main()
    {
        var variant = 7;
        var n = 19;
        var minValue = -60;
        var maxValue = 60;
        var type = "B";
        var p = 5;

        int[] testNumbers = { -10, -5, 0, 5, 10 };
        double[] empty = { };

        PrintDivider();
        PrintStudentInfo("Vlad Semeniyuk", "IPZ-21", variant);
        PrintDivider();
        var randomNumbers = CreateRandomArray(n, minValue,
maxValue, variant);

        PrintArray("Random List", randomNumbers);
        PrintDivider();
        Console.WriteLine($"Sum: {Sum(randomNumbers)}");
        PrintDivider();
        PrintDivider();
        Console.WriteLine($"Avrage: {Average(randomNumbers)}");
```

```

        PrintDivider();

        Console.WriteLine($"Min:
{FindMinimum(randomNumbers)}");

        Console.WriteLine($"Max {FindMaximum(randomNumbers)}");
        PrintDivider();

        Console.WriteLine($"Sum Divisible By For Random
Numbers: {p} = {SumDivisibleBy(randomNumbers, p)}");
        Console.WriteLine($"Sum Divisible By For Test Numbers:
{p} = {SumDivisibleBy(testNumbers, p)}");

        PrintDivider();

        Console.WriteLine($"{{CountWords("Hello World \n
kdfjkdsa \t dkfjd")}}");

        PrintDivider();


        Console.WriteLine($"Average Many: {AverageMany(9)}");
        Console.WriteLine($"Average Many: {AverageMany(9, 8,
6)}");

        Console.WriteLine($"Average Many:
{AverageMany(empty)}");

        PrintDivider();

    }

    static void PrintDivider()
    {
        const string divider =
"-----
-----";

        Console.WriteLine(divider);
    }

```

```
static void PrintStudentInfo(string fullName, string group,
int variant)
{
    Console.WriteLine($"Full Name: {fullName}");
    Console.WriteLine($"Group: {group}");
    Console.WriteLine($"Variant: {variant}");
}
```

```
static int[] CreateRandomArray(
    int length,
    int minValue,
    int maxValue,
    int seed)
{
    var result = new int[length];
    var random = new Random(seed);

    for (int i = 0; i < length; i++)
    {
        result[i] = random.Next(minValue, maxValue + 1);
    }

    return result;
}
```

```
static void PrintArray(string title, int[] numbers)
{
    Console.WriteLine($"{{title}}: [{string.Join(", ",
numbers) }]");
}
```

```
static int Sum(int[] numbers)
```

```
{  
    return numbers.Sum();  
}  
  
static double Average(int[] numbers)  
{  
    if (numbers.Length == 0) return 0;  
  
    var sum = Sum(numbers);  
    return (double)sum / numbers.Length;  
}  
  
static int FindMinimum(int[] numbers)  
{  
    var minNum = numbers[0];  
  
    for (var i = 1; i < numbers.Length; i++)  
    {  
        if (numbers[i] < minNum)  
        {  
            minNum = numbers[i];  
        }  
    }  
  
    return minNum;  
}  
  
static int FindMaximum(int[] numbers)  
{  
    var maxNum = numbers[0];
```

```
    for (var i = 1; i < numbers.Length; i++)
    {
        if (numbers[i] > maxNum)
        {
            maxNum = numbers[i];
        }
    }

    return maxNum;
}

static int SumDivisibleBy(int[] numbers, int p)
{
    var sum = 0;

    foreach (var number in numbers)
    {
        if (number % p == 0)
        {
            sum += number;
        }
    }

    return sum;
}

static int CountWords(string text)
{
    var words = RemoveExtraSpaces(text).Split(" ");
    Console.WriteLine($"Words: {string.Join(", ", words)}");
    var counter = 0;
```

```
        foreach (var word in words) counter++;

        return counter;
    }

    static string RemoveExtraSpaces(string text)
    {
        return string.Join(" ", text.Split((char[]?)null,
StringSplitOptions.RemoveEmptyEntries));
    }

    static double AverageMany(params double[] values)
    {
        if (values.Length == 0) return 0;

        double sum = 0;

        foreach (var num in values)
        {
            sum = sum + num;
        }

        return sum / values.Length;
    }
}
```

ШІ використовувався тільки для ресьорчу.